

Technology Stack Specification

Date	06 July 2026
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Team

Technology Stack Details

S.No	Architecture Layer	Technology Chosen	Justification / Purpose
1	Frontend Client-Side	/ HTML5, CSS3 (Vanilla), Vanilla JS	Sleek light-mode dashboard, responsive tabs, async Fetch API uploads, no heavy framework overhead.
2	Backend Server-Side	/ Python, Flask framework	Fast, modular routing, lightweight runtime, perfect library integration for running Python ML models.
3	Machine Learning Stack	scikit-learn, XGBoost, imbalanced-learn	High-performance classification, robust preprocessing pipelines, SMOTE class balancing, and standard scaling.
4	Data Storage	Pandas DataFrame, CSV file structures	In-memory spreadsheet processing for batch evaluations, local file output generation.
5	Version Control & CI/CD	Git, GitHub	Collaborative development, branch tracking, code layout compliance verification.
6	Cloud / Hosting	Render / public web deployment	Seamless container or script deployment for testing model predictions online.